

Unit 3, Lesson 7: Solving Problems Using Angle Pair Relationships



Benchmark

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.



Learning Target

- ☐ I can solve mathematical and real-world problems involving angle pair relationships formed by transversals and parallel lines.



3.7.1 Warm-Up: Angle Pair Relationships

- The image shows two streets intersected by a third street. Various points of interest are located at the intersections.



Match the listed points of interest with their angle relationship listed in the right column. Write the letter of the angle relationship on the line.

Points of Interest	Angle Relationship
_____ Museum and Worship Center	A. Alternate Interior Angles
_____ Bank and Day Care	B. Consecutive Interior Angles
_____ Park and Coffee Shop	C. Corresponding Angles
_____ Fire Station and Worship Center	D. Vertical Angles
_____ Community Center and Day Care	E. Alternate Exterior Angles
_____ Bank and Museum	F. Linear Pair



3.7.2 Collaboration: Solving Problems Using Angle Relationships

Work with your partner to solve each problem.

1. Mako is a steel roller coaster located at Sea World in Orlando. Some of the support beams are shown in the diagram using $\overline{KL}$, $\overline{LR}$, and $\overline{FR}$.



Determine the value of x that shows that $\overline{KL} \parallel \overline{FR}$, if $m\angle KLR = (7x - 5.6)^\circ$ and $m\angle FRL = (3x + 5.6)^\circ$.

2. Most of the streets in Miami are on a grid. Beacom Blvd. is one of the few streets that does not run north to south or east to west. The diagram shows a map of Beacom Blvd., $\overline{HG}$, between SW 4th Street, $\overline{WT}$, and SW 6th Street, $\overline{BR}$, just a few blocks from Miami Senior High School.



- a. Given that $\overline{WT} \parallel \overline{BR}$, determine the value of x if $m\angle HMB = (2x - 7)^\circ$, $m\angle WKG = (4x + 25)^\circ$, and $m\angle KMB = (6x - 29)^\circ$.

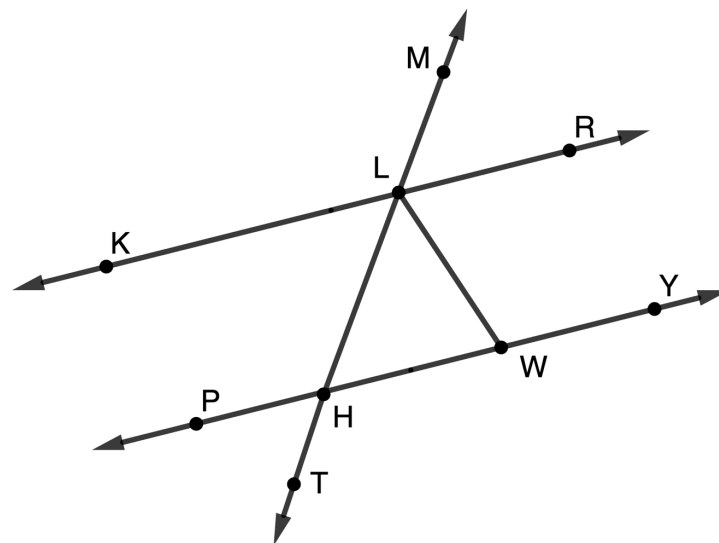
b. Determine the measure of each angle.

$m\angle HMB$

$m\angle HMR$

$m\angle TKM$

3. $\overleftrightarrow{KR}$, $\overleftrightarrow{PY}$, and $\overleftrightarrow{MT}$ are shown with $\overleftrightarrow{LW}$ where $m\angle LWH = (4y - 9\frac{1}{3})^\circ$, $m\angle WLH = (3y + 3)^\circ$, and $m\angle LHW = m\angle WLH$.



- a. Determine the value of y that shows that $\overleftrightarrow{PY} \parallel \overleftrightarrow{KR}$.

- b. Determine the measure of each angle.

$m\angle RLW$

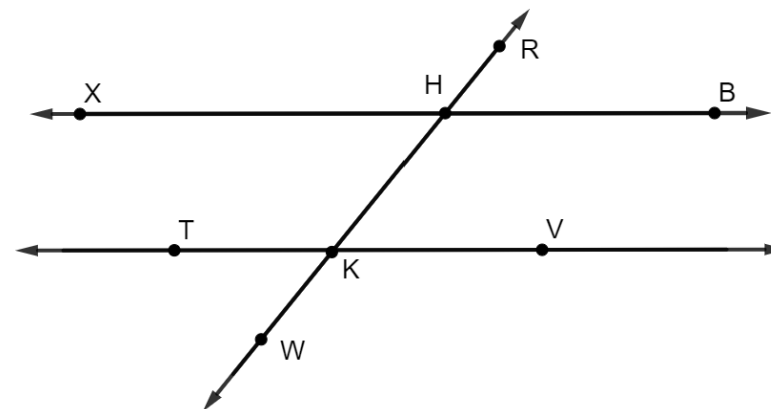
$m\angle PHT$

$m\angle KLM$



3.7.3 Your Turn!

1. $\overleftrightarrow{XB}$, $\overleftrightarrow{TV}$, and $\overleftrightarrow{RW}$ are shown. $\overleftrightarrow{XB} \parallel \overleftrightarrow{TV}$ where $\overleftrightarrow{RW}$ is a transversal with $m\angle XHR = (9x + 21)^\circ$ and $m\angle HKV = (4x + 3)^\circ$.



- a. Determine the value of x .
- b. Find the measure of each angle.

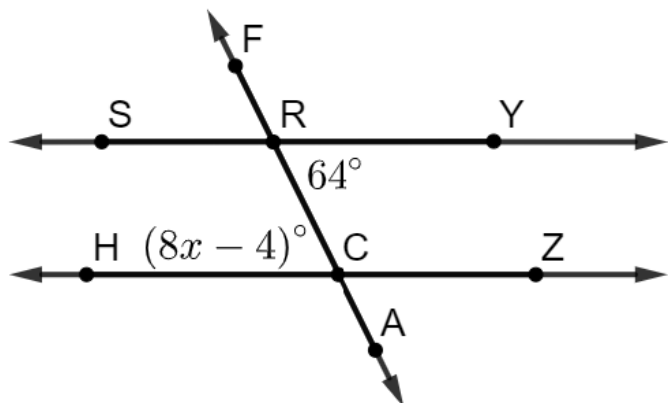
$m\angle XHR$

$m\angle HKV$

$m\angle TKW$

$m\angle BHK$

2. $\overleftrightarrow{SY} \parallel \overleftrightarrow{HZ}$, $m\angle YRC = 64^\circ$, and $m\angle RCH = (8x - 4)^\circ$.



a. Find the value of x .

b. Find the measure of each angle.

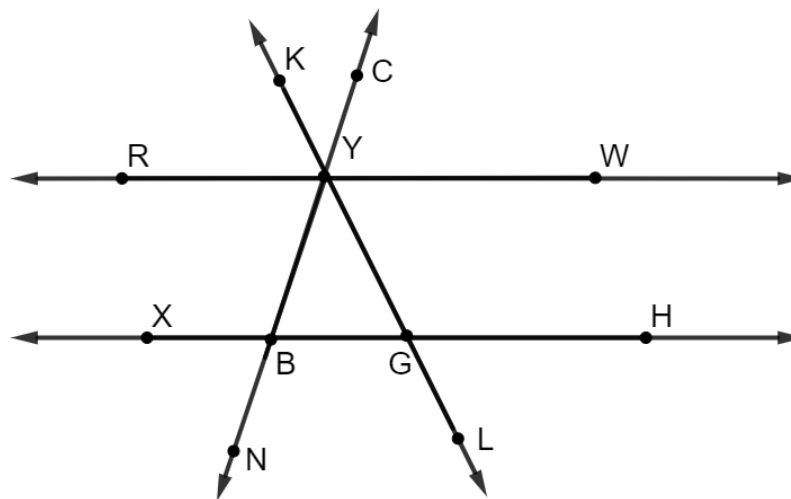
$m\angle ZCA$

$m\angle FRS$

$m\angle YRF$

$m\angle HCA$

3. $\overleftrightarrow{RW} \parallel \overleftrightarrow{XH}$, $m\angle RYK = s^\circ$, $m\angle CYW = m^\circ$, $m\angle GYB = 43^\circ$, $m\angle YBG = 72^\circ$, and $m\angle LGH = (6t + 11)^\circ$.



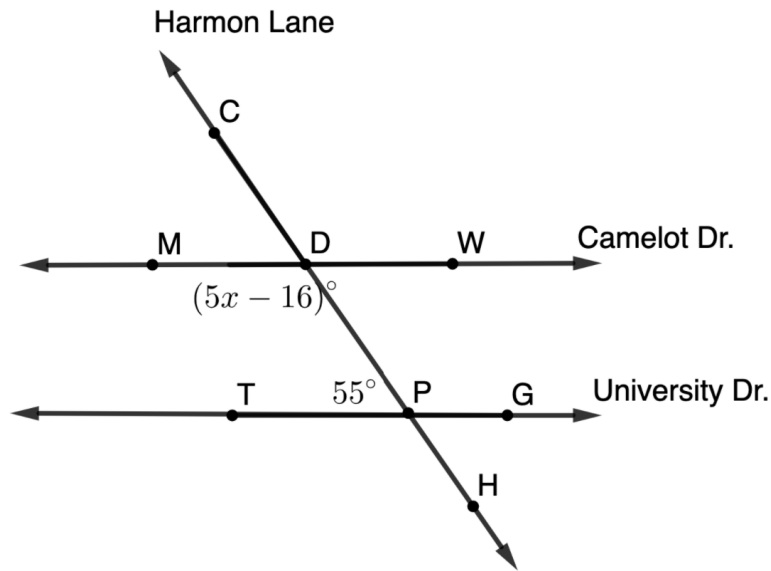
Find the value of each variable. Show your work.

t

m

s

4. Harmon Lane, $\overleftrightarrow{CH}$, crosses the streets Camelot Drive, $\overleftrightarrow{MW}$, and University Drive, $\overleftrightarrow{TG}$, as shown. $m\angle TPD = 55^\circ$, and $m\angle MDP = (5x - 16)^\circ$.



Determine the value of x that shows $\overleftrightarrow{MW} \parallel \overleftrightarrow{TG}$.



3.7.4 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current learning is related to each target.

1. I can use corresponding angles to determine angle measurements.

I need help.
I can't get started.

I'm getting there, but
I need more practice.

I understand this,
and I feel confident.



2. I can use alternate interior angles to determine angle measurements.

I need help.
I can't get started.

I'm getting there, but
I need more practice.

I understand this,
and I feel confident.



3. I can use alternate exterior angles to determine angle measurements.

I need help.
I can't get started.

I'm getting there, but
I need more practice.

I understand this,
and I feel confident.



4. I can use consecutive interior angles to determine angle measurements.

I need help.
I can't get started.

I'm getting there, but
I need more practice.

I understand this,
and I feel confident.



5. I can use consecutive exterior angles to determine angle measurements.

I need help.
I can't get started.

I'm getting there, but
I need more practice.

I understand this,
and I feel confident.

